Collective focusing in two dimensions: the central problem and verified reductions

Shuxing Yang

11 October 2026

Status. The complete two-dimensional sharp Lieb–Thirring problem is unresolved by this work. The reductions below use existing spectral theorems and standard NLS symmetries. The tensor-product bound is a direct application of a classical sharp inequality; no historical novelty is asserted. The accompanying numerical searches found no counterexample.

1 The precise target

For the two-dimensional common-density system

$$i\partial_t u_j + \Delta u_j + \rho u_j = 0, \qquad \rho = \sum_j |u_j|^2, \qquad E = \tfrac{1}{2}\sum_j \|\nabla u_j\|_2^2 - \tfrac{1}{4}\int \rho^2,$$

the eigenvalues of the full Gram matrix are conserved occupations. Let $-\Delta Q + Q - Q^3 = 0$ be the scalar ground-state equation, $M_c = \|Q\|_2^2$, and $C_1 = 2/M_c$. The question is whether, for every positive finite-trace operator Γ with finite kinetic energy,

$$\|\rho_\Gamma\|_2^2 \le C_1 \|\Gamma\|_{\mathrm{op}} \operatorname{Tr}(-\Delta\Gamma). \tag{1}$$

It is equivalent by spectral duality to

$$\operatorname{Tr}(-\Delta - V)_- \le \frac{1}{2M_c}\int_{\mathbb{R}^2} V_+^2.$$

If true, (1) yields

$$E \ge \tfrac{1}{2}(1 - \|\Gamma\|_{\mathrm{op}}/M_c)\operatorname{Tr}(-\Delta\Gamma).$$

For a fixed normalized occupation spectrum p, the total-mass threshold would therefore be $M_*(p) = M_c/p_{\max}$. Below this value all finite-component energy-space solutions are global and uniformly bounded; above it negative-energy finite-variance data exist. The latter follows by concentrating the largest mode and putting the others on disjoint broad supports. No assertion is made that all superthreshold data collapse.

2 Every failure can be sought among equal occupations

Set $m = \|\Gamma\|_{\mathrm{op}}$, $P_s = \mathbf{1}_{(s,\infty)}(\Gamma)$ and $D(P) = C_1 \operatorname{Tr}(-\Delta P) - \|\rho_P\|_2^2$. The exact identity

$$\begin{aligned} C_1 m \operatorname{Tr}(-\Delta\Gamma) - \|\rho_\Gamma\|_2^2 = m\int_0^m D(P_s)\,ds \\ + \tfrac{1}{2}\int_0^m\int_0^m \|\rho_{P_s} - \rho_{P_t}\|_2^2\,ds\,dt \end{aligned}$$

follows from $\Gamma = \int_0^m P_s\,ds$ and the Hilbert-space variance identity. The classical non-sharp kinetic inequality ensures integrability. Each P_s has finite rank. Thus a strict counterexample always yields a finite-rank projection counterexample, with all occupied modes weighted equally. This reduction does not bound its rank.

3 A failure would produce an exact collective standing wave and collapse

Write L_N for the best constant in the first-N spectral sum. Frank–Gontier–Lewin's finite-rank optimizer, bubble-decomposition and Euler–Lagrange theorems apply at $d = 2, \gamma = 1$. If $L_N > L_1 = 1/(2M_c)$ first occurs at $N = N_*$, minimality excludes splitting of maximizing sequences into smaller-rank optimizers. There is a normalized optimizer, real analytic and exponentially decaying, with

$$\int V_*^2 = 1, \qquad V_* = \frac{1}{2L_{N_*}} \sum_{j=1}^{N_*} |\phi_j|^2, \qquad (-\Delta - V_*)\phi_j = -e_j\phi_j, \qquad \langle \phi_i, \phi_j \rangle = \delta_{ij}.$$

There may be additional unoccupied negative eigenvalues; they must not be excluded by assumption.

Let $\beta = 1/(2L_{N_*}) < M_c$ and $\psi_j = \sqrt{\beta}\phi_j$. Then

$$-\Delta\psi_j - \left(\sum_k |\psi_k|^2 \right) \psi_j = -e_j\psi_j, \qquad \langle \psi_i, \psi_j \rangle = \beta\delta_{ij}. \tag{2}$$

Summing the eigenvalue equations gives $\sum_j \|\nabla\psi_j\|_2^2 = 1/2$, $\int \rho_\Psi^2 = 1$, hence $E(\Psi) = 0$. Conversely any nonzero localized solution of (2) with equal occupation $\beta < M_c$ disproves (1): the Pohozaev identity gives $E(\Psi) = 0$, and normalization of the orbitals yields the quotient $2/\beta > 2/M_c$.

Choose real eigenfunctions, let $h = T - t > 0$, and set

$$u_j(t, x) = h^{-1} \exp\left(-\frac{i|x|^2}{4h} + \frac{ie_j}{h} \right) \psi_j(x/h).$$

Pseudoconformal invariance, or direct differentiation, gives an exact NLS solution with Gram matrix βI and

$$\sum_j \|\nabla u_j(t)\|_2^2 = h^{-2} \sum_j \|\nabla\psi_j\|_2^2 + \tfrac{1}{4} \sum_j \||x|\psi_j\|_2^2.$$

Its energy is $\frac{1}{8} \sum_j \||x|\psi_j\|_2^2$. It collapses although every occupation is below M_c. These implications identify the unresolved existence question; they do not construct such a subcritical standing wave.

4 A global exclusion of complete product occupations

The classical sharp one-dimensional $\gamma = 3/2$ inequality is

$$\mathrm{Tr}(-\partial_x^2 - V)_-^{3/2} \le \tfrac{3}{16} \int V_+^2.$$

For a positive finite-rank operator A, write $B_A = \int \rho_A^2$ and $T_A = \mathrm{Tr}(-\partial_x^2 A)$. Variational duality and Schatten Hölder yield, for $a > 0$,

$$aB_A - T_A \le (\mathrm{Tr}\, A^3)^{1/3} (3/16)^{2/3} a^{4/3} B_A^{2/3}.$$

Optimization in a gives $B_A^2 \le (\operatorname{Tr} A^3)T_A/3$. For contractions $0 \le A, B \le 1$, therefore,

$$\int \rho_{A\otimes B}^2 \le \tfrac{1}{3}\sqrt{(\operatorname{Tr} A)(\operatorname{Tr} B)T_A T_B} \le \tfrac{1}{6}\operatorname{Tr}[-\Delta(A\otimes B)].$$

Convex combinations obey the same estimate by Jensen's inequality. In particular, every complete rectangular occupation $\{f_i(x)g_j(y) : 1 \le i \le m,\ 1 \le j \le n\}$ has kinetic quotient at most $1/6$, with no shape or rank restriction. The constant is attained at rank one by $\frac{1}{2}\operatorname{sech} x \operatorname{sech} y$ and approached at larger rectangular ranks by separated copies.

The strict inequality $1/6 < C_1$ has an elementary witness. For $q(x) = \operatorname{sech}|x|/\sqrt{2\pi \log 2}$ in two dimensions,

$$\frac{\int q^4}{\int |\nabla q|^2} = \frac{4\ell - 1}{2\pi\ell(2\ell+1)} > \frac{1}{6}, \qquad \ell = \log 2.$$

Indeed $69/100 < \ell < 7/10$ and $\pi < 22/7$ give the strict bound $1.76/10.56 = 1/6$. The logarithm estimates follow from $\log 2 = 2\sum_{k\ge 0}[(2k+1)3^{2k+1}]^{-1}$. Nonrectangular occupation sets and general nonseparable projections are not covered.

5 Numerical exploration and the remaining mathematical task

The recorded periodic Gaussian-well potential is

$$V(x) = A\sum_{z\in\mathbb{Z}^2} e^{-|x-2\pi z|^2/(2\sigma^2)} + c, \quad A = 2.9414883, \quad \sigma = 0.73521703, \quad c/A = 0.01513357.$$

Its computed spectral ratio is approximately 0.0425618561, below the scalar target $1/(2M_c) \simeq 0.0427317684$ by 0.398%. Plane-wave cutoffs and Bloch meshes were increased; the recorded digits were stable. Fourier and spatial evaluations of the potential cost also agree. These are floating-point checks, not rigorous enclosures. No continuum upper bound or global optimality follows.

The open task is to construct or exclude the equal-occupation collective structures in (2), without radiality, fixed-rank or single-core assumptions. The existing reductions and product bound do not accomplish this. Full derivations, source checks, code and numerical results are included in the accompanying working files.

References

[1] R. L. Frank, D. Gontier and M. Lewin, *Optimizers for the finite-rank Lieb–Thirring inequality*, arXiv:2109.05984v2 (2023), Theorem 1, Corollary 2 and Theorem 3. `https://arxiv.org/abs/2109.05984`

[2] R. L. Frank, D. Gontier and M. Lewin, *The nonlinear Schrödinger equation for orthonormal functions II. Application to Lieb–Thirring inequalities*, CMP (2021). `https://arxiv.org/abs/2002.04964`

[3] A. Laptev and T. Weidl, *Sharp Lieb–Thirring inequalities in high dimensions*, Acta Math. (2000). `https://arxiv.org/abs/math-ph/9903007`